3. $\sin^{-1}\frac{3}{5}-\cos^{-1}\frac{12}{13}+\sec^{-1}\frac{5}{3}-\operatorname{cosec}^{-1}\frac{13}{12}$ is equal to

(1) 2π (2) π (3) 0 (4) $\tan^{-1}\frac{12}{65}$

4. If $\sin^{-1}x=2\sin^{-1}\alpha$ has a solution, then

(1) $|\alpha|\le\frac{1}{\sqrt{2}}$ (2) $|\alpha|\ge\frac{1}{\sqrt{2}}$ (3) $|\alpha|<\frac{1}{\sqrt{2}}$ (4) $|\alpha|>\frac{1}{\sqrt{2}}$

5. $\sin^{-1}(\cos x)=\frac{\pi}{2}-x$ is valid for

(1) $-\pi\le x\le 0$ (2) $0\le x\le\pi$ (3) $-\frac{\pi}{2}\le x\le\frac{\pi}{2}$ (4) $-\frac{\pi}{4}\le x\le\frac{3\pi}{4}$

6. If $\sin^{-1}x+\sin^{-1}y+\sin^{-1}z=\frac{3\pi}{2}$, the value of $x^{2017}+y^{2018}+z^{2019}-\frac{9}{x^{101}+y^{101}+z^{101}}$ is

(1) 0 (2) 1 (3) 2 (4) 3

7. If $\cot^{-1}x=\frac{2\pi}{5}$ for some $x\in R$, the value of $\tan^{-1}x$ is

(1) $-\frac{\pi}{10}$ (2) $\frac{\pi}{5}$ (3) $\frac{\pi}{10}$ (4) $-\frac{\pi}{5}$

8. The domain of the function defined by $f(x)=\sin^{-1}\sqrt{x-1}$ is

(1) $[1,\ 2]$ (2) $[-1,\ 1]$ (3) $[0,\ 1]$ (4) $[-1,\ 0]$

9 If $x=\frac{1}{5}$, the value of $\cos\left(\cos^{-1}x+2\sin^{-1}x\right)$ is

(1) $-\sqrt{\frac{24}{25}}$ (2) $\sqrt{\frac{24}{25}}$ (3) $\frac{1}{5}$ (4) $-\frac{1}{5}$

10. $\tan^{-1}\left(\frac{1}{4}\right)+\tan^{-1}\left(\frac{2}{9}\right)$ is equal to

(1) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ (2) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ (3) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ (4) $\tan^{-1}\left(\frac{1}{2}\right)$

11. If the function $f(x)=\sin^{-1}\left(x^2-3\right)$, then x belongs to

(1) $[-1, 1]$ (2) $\left[\sqrt{2}, 2\right]$

(3) $\left[-2, -\sqrt{2}\right]\cup\left[\sqrt{2},\ 2\right]$ (4) $\left[-2, -\sqrt{2}\right]$

12. If $\cot^{-1}2$ and $\cot^{-1}3$ are two angles of a triangle, then the third angle is

(1) $\frac{\pi}{4}$ (2) $\frac{3\pi}{4}$ (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{3}$

13. $\sin^{-1}\left(\tan\frac{\pi}{4}\right)-\sin^{-1}\left(\sqrt{\frac{3}{x}}\right)=\frac{\pi}{6}$. Then x is a root of the equation

(1) $x^2-x-6=0$ (2) $x^2-x-12=0$ (3) $x^2+x-12=0$ (4) $x^2+x-6=0$

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